

AIRTEL NETWORK PERFORMANCE & CUSTOMER EXPERIENCE ANALYTICS

SQL & Power BI Data Analytics Project

FINAL PBIX-ALIGNED PROJECT DOCUMENTATION

Strict visual-by-visual alignment with Airtel_Network_Analytics.pbix

1. Project Title

Airtel Network Performance & Customer Experience Analytics

2. Industry

Telecommunication Industry. The project analyzes network performance, customer complaints, customer satisfaction, location-wise performance, and customer usage patterns.

Dataset note: The dataset is a synthetic/project dataset created for learning and portfolio purposes, not Airtel's confidential operational data.

3. Problem Statement

Telecommunication companies generate large volumes of network and customer data. It can be difficult to identify poor-performing locations, network quality issues, customer complaints, and factors affecting customer satisfaction. This project uses SQL and Power BI to identify performance problems and provide actionable insights.

Identify poor network availability and slow internet speed.

Analyze call-drop rate, latency, and network downtime.

Track customer complaints and complaint resolution.

Measure customer satisfaction and NPS.

Compare 4G and 5G performance.

Analyze state/city performance and customer usage patterns.

4. Objectives

Analyze overall network performance.

Compare 4G and 5G network KPIs.

Identify locations with poor network quality.

Analyze customer complaints and complaint categories.

Measure customer satisfaction and NPS.

Evaluate complaint resolution performance.

Understand customer data and voice usage.

Create interactive Power BI dashboards.

Provide data-driven business recommendations.

5. Dataset Description

The project contains five primary data tables with 10,000 rows per table, giving 50,000 rows across the five source tables.

Key fields include Customer_ID, dates, State, City, Network_Type, Customer_Segment, network availability, speed, latency, call-drop rate, complaints, satisfaction, NPS, data usage and voice usage.

6. Database / Power BI Model Design

The model contains five primary source tables. Customer_ID is the common key used to connect customer-related records. The PBIX also contains a Date_Table and an All measures table.

Customer_Data - customer master information.

Network_Performance - network performance records.

Customer_Experience - complaints and customer experience metrics.

Location_Analysis - state and city performance metrics.

Usage_Analysis - customer data and voice usage.

Date_Table - report date/month dimension used by date slicers and monthly visuals.

All measures - dedicated Power BI measure table containing the report's DAX measures.

Relationship concept: Customer_Data → Customer_ID → Network_Performance / Customer_Experience / Location_Analysis / Usage_Analysis.

7. SQL Queries

Total Customers

```
SELECT COUNT(DISTINCT Customer_ID) AS Total_Customers
FROM Customer_Data;
```

Total Complaints

```
SELECT COUNT(*) AS Total_Complaints
FROM Customer_Experience
WHERE Complaint_Flag = 1;
```

Average Network Availability

```
SELECT ROUND(AVG(Network_Availability_Pct),2) AS Avg_Network_Availability
FROM Network_Performance;
```

Average Download Speed

```
SELECT ROUND(AVG(Download_Speed_Mbps),2) AS Avg_Download_Speed  
FROM Network_Performance;
```

4G vs 5G Performance

```
SELECT Network_Type,  
ROUND(AVG(Download_Speed_Mbps),2) AS Avg_Download_Speed,  
ROUND(AVG(Latency_ms),2) AS Avg_Latency,  
ROUND(AVG(Call_Drop_Rate_Pct),2) AS Avg_Call_Drop  
FROM Network_Performance  
GROUP BY Network_Type;
```

State-wise Performance

```
SELECT State,  
ROUND(AVG(Network_Availability_Pct),2) AS Avg_Availability,  
ROUND(AVG(Download_Speed_Mbps),2) AS Avg_Speed,  
ROUND(AVG(Call_Drop_Rate_Pct),2) AS Avg_Call_Drop  
FROM Location_Analysis  
GROUP BY State  
ORDER BY Avg_Availability DESC;
```

Complaint Categories

```
SELECT Complaint_Category, COUNT(*) AS Complaint_Count  
FROM Customer_Experience  
WHERE Complaint_Flag = 1  
GROUP BY Complaint_Category  
ORDER BY Complaint_Count DESC;
```

8. Exact Power BI Report Structure

The PBIX contains exactly 4 report pages, in this order:

1. 1.Executive Summary
2. 2.Network Performance
3. 3.Customer Experience
4. 4.Usage Analysis & Location Analysis

Each report page is configured as Fit to Page at 1280 × 720. The following sections document the actual visual types, titles, fields, measures, filters, and navigation elements found in the PBIX metadata.

9.1 Page 1 - 1.Executive Summary

Exact PBIX page name: 1.Executive Summary

Page size: 1280 × 720. Display mode: FitToPage.

Page inventory: 5 data charts, 7 card visuals, 5 slicers, 1 page navigator, 0 action button(s), 7 textboxes, 7 image(s), and 6 shape(s).

Filters / slicers:

customer_data.State

customer_data.Network_Type

customer_data.City

Date_Table.Date

customer_data.Customer_Segment

Visual-by-visual inventory:

Named textbox labels exactly present in the PBIX:

Call Drop Rate

Total_coustomer

Avg Download Speed

Total complaints

Customer Satisfaction

Avg Network Availability

9.2 Page 2 – 2.Network Performance

Exact PBIX page name: 2.Network Performance

Page size: 1280 × 720. Display mode: FitToPage.

Page inventory: 6 data charts, 5 card visuals, 5 slicers, 1 page navigator, 2 action button(s), 1 textboxes, 5 image(s), and 0 shape(s).

Filters / slicers:

customer_data.State

customer_data.Customer_Segment

Date_Table.Date

customer_data.City

customer_data.Network_Type

Visual-by-visual inventory:

Named textbox labels exactly present in the PBIX:

2.NETWORK PERFORMANCE

9.3 Page 3 – 3.Customer Experience

Exact PBIX page name: 3.Customer Experience

Page size: 1280 × 720. Display mode: FitToPage.

Page inventory: 5 data charts, 6 card visuals, 5 slicers, 1 page navigator, 2 action button(s), 1 textboxes, 6 image(s), and 0 shape(s).

Filters / slicers:

customer_data.City

Date_Table.Date

customer_data.State

customer_data.Customer_Segment

customer_data.Network_Type

Visual-by-visual inventory:

Named textbox labels exactly present in the PBIX:

3.CUSTOMER EXPERIENCE

9.4 Page 4 – 4.Usage Analysis & Location Analysis

Exact PBIX page name: 4.Usage Analysis & Location Analysis

Page size: 1280 × 720. Display mode: FitToPage.

Page inventory: 5 data charts, 6 card visuals, 5 slicers, 1 page navigator, 1 action button(s), 1 textboxes, 6 image(s), and 0 shape(s).

Filters / slicers:

customer_data.City

customer_data.Network_Type

customer_data.State

customer_data.Customer_Segment

Date_Table.Date

Visual-by-visual inventory:

Named textbox labels exactly present in the PBIX:

4.Usage&Location; Analysis

10. Exact Power BI Measures

The PBIX uses 18 distinct measures from the 'All measures' table. The names below are taken directly from visual query references in the report.

Strict-alignment note: this document records the measure names and their actual report usage. It does not invent DAX formulas that are not exposed in the report

definition metadata.

11. Visual Summary by Page

12. Filters, Slicers and Navigation

The same core slicer set is used across the report pages, with fields taken from the customer and date dimensions.

Page navigation: each of the four report pages contains one pageNavigator visual.

Action buttons: Customer Experience contains two actionButton visuals; Network Performance contains two; Usage Analysis & Location Analysis contains one. Executive Summary contains no actionButton visual.

13. Project Workflow

Collect / prepare the synthetic project dataset.

Load the five primary tables into MySQL for SQL analysis.

Clean and validate data fields and date values.

Create relationships and the Date_Table in Power BI.

Create DAX measures in the All measures table.

Build the four report pages using the exact fields and measures documented above.

Apply slicers and page navigation.

Use SQL outputs and Power BI visuals to identify network and customer-experience patterns.

Communicate business insights and recommendations.

14. Business Insights

Identify states and cities with low network availability.

Identify locations with high call-drop rates and downtime.

Compare 4G and 5G speed, latency, and call-drop performance.

Identify the most common customer complaint categories.

Track complaint volume and resolution performance.

Identify customer segments with lower satisfaction.

Understand data and voice usage by segment and location.

Use monthly trends to detect changes in network and customer experience.

15. Recommendations

Prioritize network infrastructure improvements in low-availability locations.

Investigate and optimize high call-drop areas.

Improve capacity in locations with consistently low internet speed.

Reduce complaint resolution time by monitoring pending cases.

Focus customer-experience initiatives on high-volume complaint categories.

Use 4G/5G comparisons to prioritize network upgrades.

Monitor monthly KPIs through Power BI for early issue detection.

Review state and city performance regularly for data-driven network planning.

16. Final Strict Alignment Audit

17. Conclusion

This final document is aligned to the uploaded Airtel_Network_Analytics.pbix report definition. It preserves the original project scope and SQL documentation while adding a strict visual-by-visual inventory of the four Power BI pages, slicers, navigation elements, chart titles, visual types, fields, and measures. The document is intended to be used together with the PBIX as the final project documentation.